From Refining Capacity to Import Dependence: Portugal's Diesel and Gasoline Market, 2005–2024

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Abstract

Portugal's refining system gained diesel conversion capability when the Sines hydrocracker entered commercial production in 2013, and lost one of its two refineries when refining ceased at Matosinhos in May 2021. Using monthly JODI physical flows, the Eurostat oil balance for Portugal and Spain, DGEG national statistics and the European Commission Weekly Oil Bulletin, this paper measures how the product balance and price exposure changed over 2005–2024.

The 2013 upgrade produced an unambiguous structural break. Diesel refinery output, exports and the net-import-to-demand ratio all break at 2013 and survive Benjamini–Hochberg adjustment across sixteen tests, and Portugal moved from net diesel importer to net diesel exporter. After May 2021 the direction reverses: by 2023 domestic manufacturing covered 0.66 of diesel demand, the lowest value in the twenty-year window, while imports reached a series high of 1,597 kt.

Two annual designs disagree about 2022, and the disagreement is diagnostic rather than troubling. Chow tests detect nothing on three post-event observations; interrupted-trend models on the same series detect a level shift of $-1{,}783$ kt in diesel exports ($p < 0.001$). A monthly segmented model with month fixed effects separates the May 2021 closure from the March 2022 energy shock and finds both: diesel refinery output falls 105 kt per month at the closure boundary and a further 152 kt per month during the stress period, with the net-import-to-demand ratio rising 0.20 and 0.38 respectively.

Spanish diesel output covered 0.85–0.93 of Spanish demand throughout, with no trend, while Portugal's fell to 0.68. A shock common to both countries cannot account for a divergence that appears in only one. On prices, short-run pass-through from Spanish to Portuguese pre-tax diesel prices rises from 0.73 to approximately 1.01; gasoline contemporaneous pass-through does not move, but its error-correction speed roughly triples, so both fuels became more tightly linked to Spain through different channels.

The paper reports associations around documented events. It does not claim a causal effect of the closure: the transition and the largest European energy shock in decades fall fourteen months apart, and no design here separates their contributions.

Contents

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1 Introduction and Research Question

Portugal operated two refineries for most of the study period. Sines, the larger, received a hydrocracker that entered commercial production in January 2013 and was intended to raise diesel yield relative to fuel oil. Matosinhos ceased refining in May 2021 following a December 2020 announcement, concentrating Portuguese refining at Sines. Fourteen months later, in March 2022, the invasion of Ukraine and the subsequent withdrawal from Russian oil products disturbed European product supply generally, and vacuum gas oil, a diesel feedstock at Sines, in particular.

The empirical question is whether the loss of refining capacity increased Portugal's dependence on imported finished petroleum products and reduced supply resilience, and whether Portugal–Spain price co-movement changed after the closure.

The paper does not assume that closure raised prices. It separates four mechanisms and tests each in turn:

$$\text{refining capability} \rightarrow \text{production and trade balance} \rightarrow \text{physical import dependence} \rightarrow \text{price co-movement.}$$

Four hypotheses were pre-specified before estimation:

1. the 2013 hydrocracker changed diesel-producing capability and the trade balance;
2. the 2021 Matosinhos transition changed import dependence and domestic output coverage;
3. 2022 is a distinct supply-system stress episode rather than clean post-treatment evidence;
4. any price change must be evaluated on pre-tax prices, since tax changes would otherwise be read as pass-through.

Throughout, four evidence levels are kept distinct: a *documented event*, a *descriptive change*, a *statistical association or structural break*, and a *causal effect*. Only the first three are claimed anywhere in this paper.

2 Data

2.1 Sources

Four independent bodies publish the series used here. Their identifiers, roles and retrieval points were frozen at the time of extraction; full URLs appear in the References.

- **JODI Oil World Database**: monthly secondary-product imports, exports, refinery output and demand. This is the main physical-flow source because it permits transparent time aggregation and carries per-observation assessment codes.
- **Eurostat `nrg_cb_oil`**: annual oil supply, transformation and consumption for Portugal and Spain. This is the only source covering both countries on a common definition.
- **DGEG**, the Portuguese national statistical authority: annual product trade workbooks and a long domestic-sales series.
- **European Commission Weekly Oil Bulletin**: weekly consumer prices with and without taxes from 2005.

Galp corporate disclosures supply dated refinery events. They are used as event metadata only, never as outcome data, and the announcement date of December 2020 is deliberately not treated as the physical closure date.

2.2 Coverage

Table 1 states what each source covers over the study window.

Table 1: Source coverage, 2005–2024.

Source	**Role**	**Coverage**
JODI	Monthly physical flows	Complete. 160 of 160 annual product-flow cells; every year carries 12 observed months and assessment code 1
Eurostat	Annual balance, PT and ES	Complete for both countries, 160 of 160 cells each
DGEG trade	National trade statistics	**2019–2024 only.** Earlier annual workbooks are not published in this series
DGEG sales	Domestic market sales	Complete, 1970–2024
EC Bulletin	Weekly consumer prices	Complete. 995 Portuguese and 994 Spanish weekly observations per product, 2005-01-03 to 2024-12-30

The monthly panel spans 2002-01 to 2026-05 for both products, of which 293 months per product fall inside the modelling window.

2.3 The one material coverage gap

DGEG publishes annual product trade workbooks only from 2019. For 2005–2018 the trade cross-check therefore runs against Eurostat rather than against the DGEG primary publication. This is weaker corroboration than it appears, because Eurostat's Portuguese oil data is compiled from the DGEG national submission and tracks it to a median difference of 0.00% across the 2019–2024 cells where both exist. Eurostat is best understood as a second channel for the same national statistics, not as an independent third opinion.

Domestic demand is unaffected. The DGEG long sales workbook covers the whole window, so demand carries three sources throughout.

2.4 Product definitions

Diesel is JODI `GASDIES` and Eurostat `O4671`, gas oil and diesel oil. Gasoline is JODI `GASOLINE` and Eurostat `O4652`, motor gasoline. Aviation gasoline is excluded from both. In the DGEG sales workbook, gasoline is the *Gasolinas* total, and diesel requires coloured and marked gasoil to be added to road gasoil; road gasoil alone sits about ten per cent below the JODI and Eurostat demand concept. Comparability is audited against a published product crosswalk, which flags the DGEG trade labels as requiring review before numerical reconciliation.

Differing biofuel treatment does not drive residual differences between sources. The Eurostat blended-biofuel wedge, `O4671` less `O4671XR5220B`, is 0.0 kt on exports in every year of the window and at most 4.5% on imports, so blending cannot explain any export divergence at all.

2.5 Source reconciliation

JODI and DGEG are the only genuinely independent pair. Across the 24 overlapping 2019–2024 trade cells, 66.7% agree within tolerance. Over the full window, JODI and Eurostat agree on 89% of 80 cells, with only three of the nine divergences occurring before 2019.

A cell is flagged only when it breaches both the absolute and the percentage tolerance. Applying them as alternatives makes a 25 kt floor bind on series of roughly 1,600 kt, an effective tolerance of 1.6%, which would flag agreement as close as 2.1% as failure. Each of the eight remaining divergent cells is enumerated in the analysis configuration, together with the series Eurostat identifies as the outlier: six where JODI is out of line, two where the DGEG workbook is. An unlisted divergence fails the readiness gate however few cells it touches.

3 Methods

For fuel j in year t, with imports M, exports X, domestic demand D and refinery output Q^{refinery}:

$$\text{NetImports}_{j,t} = M_{j,t} - X_{j,t}, \qquad \text{GrossImportDependence}_{j,t} = \frac{M_{j,t}}{D_{j,t}},$$

$$\text{NetImportToDemandRatio}_{j,t} = \frac{M_{j,t} - X_{j,t}}{D_{j,t}}, \qquad \text{RefineryOutputToDemandRatio}_{j,t} = \frac{Q^{\text{refinery}}_{j,t}}{D_{j,t}}.$$

The refinery-output-to-demand ratio is *not* self-sufficiency. Portuguese refinery output may be exported while domestic demand is met partly by imports, and for gasoline that is the normal state of the system throughout the window.

Annual breaks. Chow tests on a linear trend at pre-specified break years, with 2021 excluded as a transition year so that it is assigned to neither side, and Benjamini–Hochberg adjustment across all sixteen tests.

Annual interrupted trends. Ordinary least squares with a common trend, a post-event level term and a post-event slope term, HAC(1) covariance, 2021 again excluded. This design retains all nineteen observations rather than splitting the series.

Monthly event timing. A segmented regression on 293 monthly observations with month fixed effects, a calendar-month trend and HAC(3) errors. Each phase-trend interaction is centred on its phase boundary, so a phase coefficient is the level shift at that boundary against the extrapolated pre-closure trend, and the companion trend term is the change in slope within the phase. The two must be quoted together.

Prices. Model family is selected from stationarity and cointegration diagnostics rather than assumed. HAC(8) covariance throughout. Pre-tax prices are the primary measure.

4 Descriptive Evidence

4.1 Summary statistics

Over 2005–2024, Portuguese diesel demand averaged 5,064 kt against average refinery output of 4,745 kt, average imports of 1,020 kt and average exports of 849 kt. Gasoline averaged demand of 1,230 kt against output of 2,272 kt, imports of 178 kt and exports of 1,249 kt. The asymmetry is structural: Portugal consumes roughly four times as much diesel as gasoline while manufacturing roughly twice as much gasoline as it consumes.

Peaks are informative about timing. Diesel refinery output and exports both peak in 2015, two years after the hydrocracker. Diesel imports peak in 2023, two years after the closure. Gasoline imports also peak in 2023. Diesel and gasoline demand both peak in 2005, at the start of the window.

4.2 Diesel

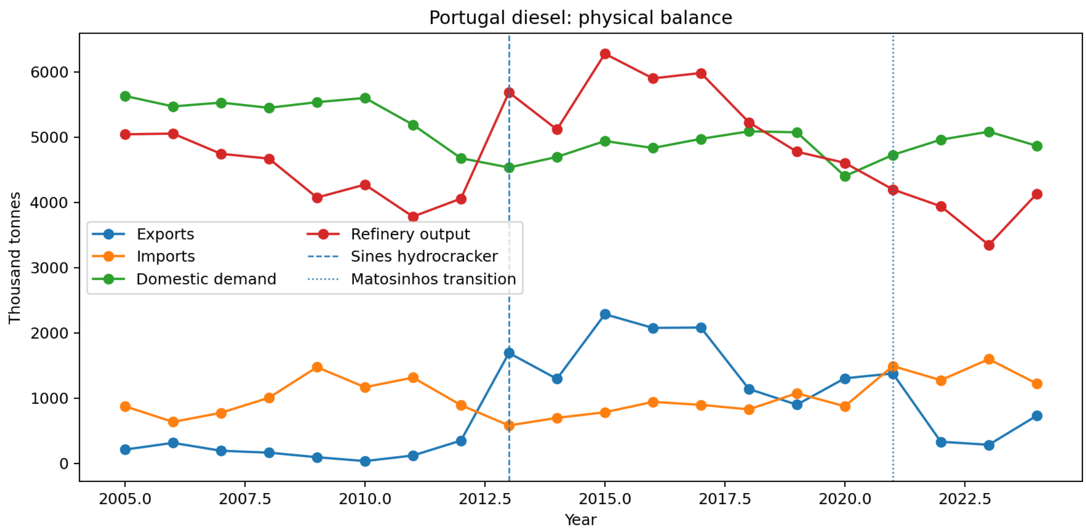

Figure 1: Portuguese diesel imports, exports, refinery output and demand, 2005–2024.

Table 2 gives the annual diesel panel. Three regimes are visible. From 2005 to 2012 Portugal is a net diesel importer with output covering 73–92 per cent of demand. From 2013 the ratio exceeds one and the net-import-to-demand ratio turns negative: Portugal manufactures more diesel than it consumes and exports the surplus. From 2021 the system returns to net import dependence, and by 2023 output covers only 66 per cent of demand.

4.3 Gasoline

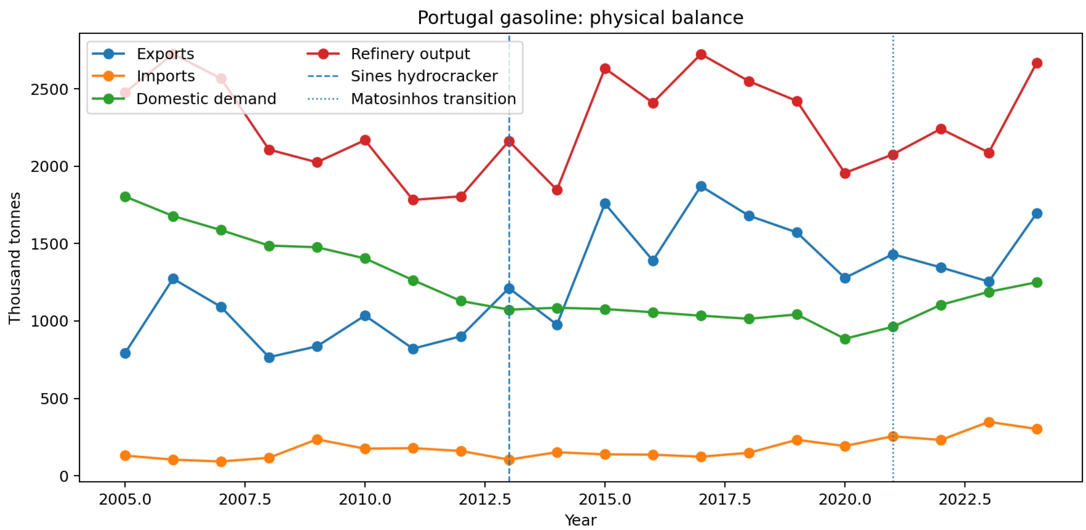

Figure 2: Portuguese gasoline imports, exports, refinery output and demand, 2005–2024.

Gasoline behaves differently in every year (Table 3). Portugal is a net gasoline exporter throughout, with output covering between 1.37 and 2.63 times domestic demand. There is no regime change comparable to diesel's: the ratio rises after 2013 and drifts down after 2021, but never approaches one. This is precisely the asymmetry the ratio definition warns about. A system can manufacture a large surplus of one light product while importing another, and reading the ratio as self-sufficiency would obscure that.

Table 2: Portugal diesel annual balance, kt and ratios.

Year	Imports	Exports	Demand	Refinery out.	Gross imp. dep.	Net imp./dem.	Ref. out./dem.
2005	875	211	5,632	5,043	0.16	0.12	0.90
2006	635	314	5,471	5,055	0.12	0.06	0.92
2007	775	193	5,530	4,745	0.14	0.11	0.86
2008	1,006	164	5,450	4,671	0.18	0.15	0.86
2009	1,475	94	5,537	4,074	0.27	0.25	0.74
2010	1,167	35	5,601	4,273	0.21	0.20	0.76
2011	1,315	120	5,191	3,782	0.25	0.23	0.73
2012	892	349	4,678	4,059	0.19	0.12	0.87
2013	581	1,692	4,535	5,687	0.13	−0.24	1.25
2014	697	1,295	4,697	5,117	0.15	−0.13	1.09
2015	783	2,286	4,941	6,279	0.16	−0.30	1.27
2016	942	2,076	4,834	5,901	0.19	−0.23	1.22
2017	897	2,082	4,974	5,984	0.18	−0.24	1.20
2018	827	1,140	5,089	5,228	0.16	−0.06	1.03
2019	1,075	899	5,075	4,777	0.21	0.03	0.94
2020	875	1,304	4,403	4,606	0.20	−0.10	1.05
2021	1,491	1,378	4,728	4,199	0.32	0.02	0.89
2022	1,275	331	4,962	3,941	0.26	0.19	0.79
2023	1,597	285	5,084	3,348	0.31	0.26	0.66
2024	1,222	731	4,868	4,133	0.25	0.10	0.85

Table 3: Portugal gasoline annual balance, kt and ratios.

Year	Imports	Exports	Demand	Refinery out.	Gross imp. dep.	Net imp./dem.	Ref. out./dem.
2005	130	792	1,804	2,476	0.07	−0.37	1.37
2006	104	1,273	1,678	2,726	0.06	−0.70	1.62
2007	92	1,091	1,587	2,566	0.06	−0.63	1.62
2008	116	766	1,487	2,107	0.08	−0.44	1.42
2009	235	836	1,476	2,025	0.16	−0.41	1.37
2010	175	1,035	1,404	2,168	0.12	−0.61	1.54
2011	178	821	1,264	1,782	0.14	−0.51	1.41
2012	160	901	1,129	1,805	0.14	−0.66	1.60
2013	104	1,212	1,073	2,162	0.10	−1.03	2.01
2014	152	976	1,085	1,847	0.14	−0.76	1.70
2015	138	1,758	1,077	2,632	0.13	−1.50	2.44
2016	136	1,390	1,056	2,410	0.13	−1.19	2.28
2017	123	1,871	1,034	2,724	0.12	−1.69	2.63
2018	148	1,680	1,014	2,549	0.15	−1.51	2.51
2019	232	1,572	1,042	2,421	0.22	−1.29	2.32
2020	192	1,278	885	1,956	0.22	−1.23	2.21
2021	255	1,431	962	2,076	0.27	−1.22	2.16
2022	231	1,346	1,104	2,241	0.21	−1.01	2.03
2023	348	1,254	1,188	2,088	0.29	−0.76	1.76
2024	302	1,697	1,250	2,669	0.24	−1.12	2.14

4.4 Dependence ratios

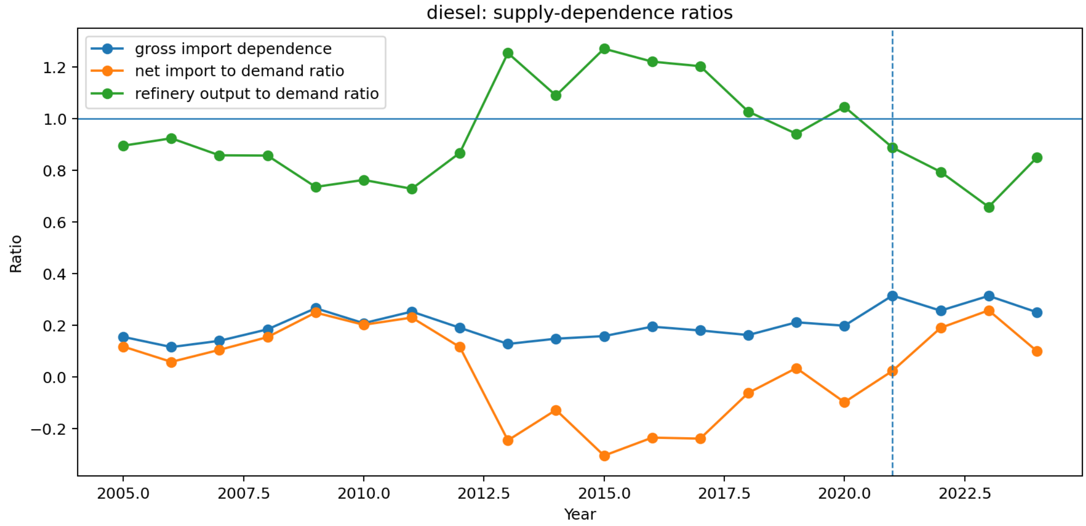

Figure 3: Diesel gross import dependence, net imports to demand, and refinery output to demand. The output ratio crossing one in 2013 and falling back after 2021 is the central physical result of this paper.

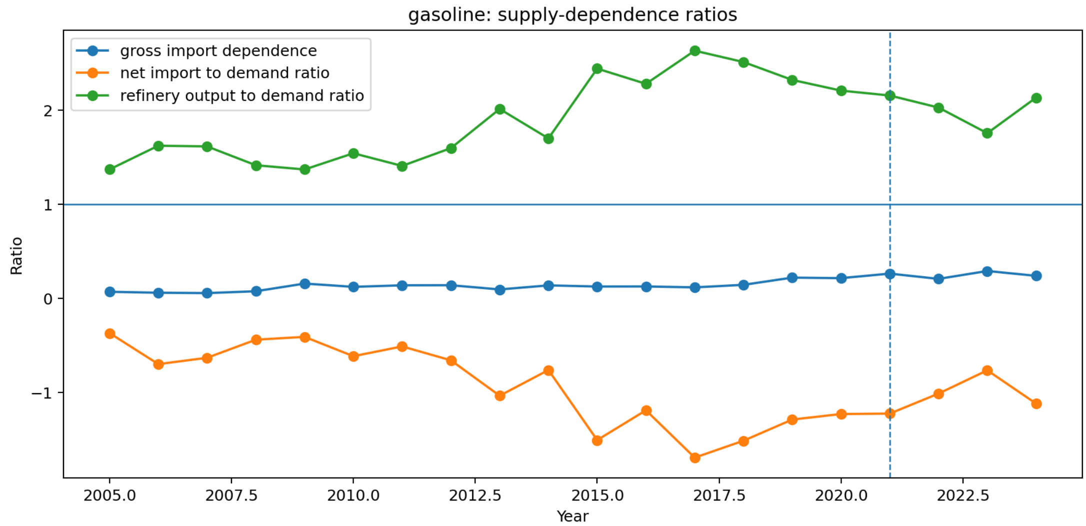

Figure 4: Gasoline dependence ratios. The output ratio remains far above one throughout, so no regime change comparable to diesel occurs.

4.5 Event years

Table 4 extracts the years that bracket each documented event, together with the refining regime in force. The 2021 row is a transition year and is excluded from the annual before/after designs.

Table 4: Balance at the event years.

Year	Product	Exports	Imports	Demand	Ref. out.	Net imp./dem.	Regime
2012	diesel	349	892	4,678	4,059	+0.116	two refineries
2013	diesel	1,692	581	4,535	5,687	−0.245	two refineries
2020	diesel	1,304	875	4,403	4,606	−0.097	two refineries
2021	diesel	1,378	1,491	4,728	4,199	+0.024	transition
2022	diesel	331	1,275	4,962	3,941	+0.190	Sines only
2023	diesel	285	1,597	5,084	3,348	+0.258	Sines only
2024	diesel	731	1,222	4,868	4,133	+0.101	Sines only
2012	gasoline	901	160	1,129	1,805	−0.656	two refineries
2013	gasoline	1,212	104	1,073	2,162	−1.033	two refineries
2020	gasoline	1,278	192	885	1,956	−1.227	two refineries
2021	gasoline	1,431	255	962	2,076	−1.222	transition
2022	gasoline	1,346	231	1,104	2,241	−1.010	Sines only
2023	gasoline	1,254	348	1,188	2,088	−0.763	Sines only
2024	gasoline	1,697	302	1,250	2,669	−1.116	Sines only

5 Structural Breaks

5.1 Chow tests

Table 5 reports the Chow tests that reach nominal significance; the remaining eleven of sixteen have $p > 0.11$. Three survive Benjamini–Hochberg adjustment comfortably. All three are diesel outcomes at 2013, and all move in the direction the engineering implies: more diesel manufactured, more exported, and a net-import position that turns negative.

Table 5: Chow tests at pre-specified break years, 2021 excluded as a transition year.

Product	Outcome	Break	F	p	BH p	$n_{\text{pre}}/n_{\text{post}}$
diesel	net imports / demand	2013	19.62	0.0002	0.0026	8/8
diesel	refinery output	2013	16.68	0.0003	0.0027	8/8
diesel	exports	2013	13.79	0.0008	0.0041	8/8
gasoline	refinery output	2013	5.23	0.0232	0.0867	8/8
diesel	imports	2013	4.95	0.0271	0.0867	8/8

No 2022 test reaches significance; the smallest adjusted p is 0.378, on eight pre- and three post-event years.

5.2 Interrupted trends, and why they disagree about 2022

The absence of a detected 2022 Chow break is specific to that test. A Chow test asks whether two independently fitted trends beat one; with three post-event annual observations it cannot answer that question. An interrupted-trend model asks whether the level shifts against a trend estimated from the whole series, a far weaker demand on the post-event data. On that specification, reported in Table 6, 2022 is large and precisely estimated for both products.

The two designs are not in conflict about the world; they are in conflict about what three annual observations can support. Any statement that annual evidence detects no 2022 break should therefore be read as specific to the Chow specification. The interrupted-trend models and the monthly design of Section 6 agree with each other in sign and rough magnitude.

Table 6: Interrupted-trend models, annual series, HAC(1), $n = 19$, 2021 excluded.

		Level change		Slope change	
Product	Outcome, event	Estimate	p	Estimate	p
diesel	exports, 2013	+1,900.2	< 0.001	−139.9	< 0.001
diesel	exports, 2022	−1,782.9	< 0.001	+79.9	0.206
diesel	net imp./dem., 2013	−0.520	< 0.001	+0.030	0.004
diesel	net imp./dem., 2022	+0.457	< 0.001	−0.021	0.441
gasoline	exports, 2013	+537.8	0.001	+31.5	0.271
gasoline	exports, 2022	−447.5	0.020	+122.8	0.049
gasoline	net imp./dem., 2013	−0.676	0.001	+0.027	0.328
gasoline	net imp./dem., 2022	+0.732	< 0.001	+0.024	0.728

6 Monthly Event Analysis

6.1 Phase means

The monthly panel is partitioned into four phases: before May 2021, the Matosinhos transition from May 2021 to February 2022, the energy-stress period from March 2022 to December 2022, and post-stress from January 2023. Table 7 gives diesel phase means.

Table 7: Diesel monthly means by event phase, kt per month unless noted.

Phase	Months	Imports	Exports	Ref. out.	Net imp./dem.
Pre-closure	232	75.4	65.4	411.5	0.013
Matosinhos transition	10	133.3	63.0	321.9	0.162
Energy stress 2022	10	111.1	27.7	324.6	0.193
Post-stress	41	120.1	35.1	300.8	0.210

Mean monthly diesel imports rise by roughly three quarters between the pre-closure phase and every subsequent phase, refinery output falls by a fifth and then a quarter, and the net-import-to-demand ratio moves from approximately zero to 0.21. The phase means are descriptive and take no account of seasonality or trend; the model below does both.

6.2 Segmented model

Both phases carry independent, statistically distinguishable effects, which is the reason a monthly design was required at all: annual data cannot separate a May 2021 event from a March 2022 one. Diesel refinery output falls by roughly 105 kt per month at the closure boundary and by a further 152 kt per month in the stress period. The net-import-to-demand ratio rises by 0.20 at the closure and by 0.38 during the stress phase, and the closure phase also carries a significant upward slope of 0.036 per month, meaning dependence continued to widen within that phase rather than stepping once and holding.

7 The 2022 Stress Episode

Against a 2013–2019 baseline, 2022 diesel exports of 331 kt sit 2.44 standard deviations below the mean and at the zeroth percentile of the baseline distribution, while diesel imports of 1,275 kt sit 2.74 standard deviations above and at the hundredth percentile. Robust statistics agree: the robust z-score for diesel imports is 2.63 against the baseline median and median absolute deviation.

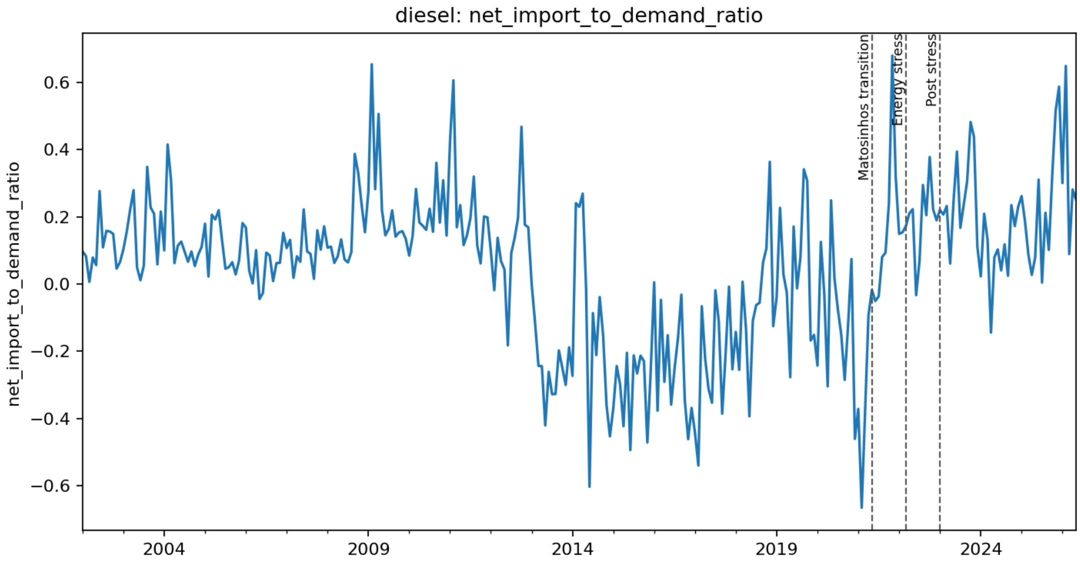

Figure 5: Monthly diesel net imports to demand, with the May 2021, March 2022 and January 2023 phase boundaries marked.

Table 8: Segmented monthly event model, diesel, $n = 293$ months, month fixed effects, HAC(3). A phase coefficient is the level shift at that phase boundary against the extrapolated pre-closure trend.

Outcome	Term	Estimate	Std. error	p
Refinery output (kt/month)	Matosinhos transition	−105.21	36.02	0.004
	phase trend	−3.47	6.62	0.601
	Energy stress 2022	−151.77	23.31	< 0.001
	phase trend	+5.88	3.51	0.094
Imports (kt/month)	Matosinhos transition	+42.23	23.40	0.071
	phase trend	+2.75	6.17	0.656
	Energy stress 2022	+28.36	11.22	0.012
	phase trend	+1.08	2.53	0.670
Net imports / demand	Matosinhos transition	+0.2005	0.0713	0.005
	phase trend	+0.0360	0.0153	0.019
	Energy stress 2022	+0.3755	0.0554	< 0.001
	phase trend	+0.0083	0.0071	0.244

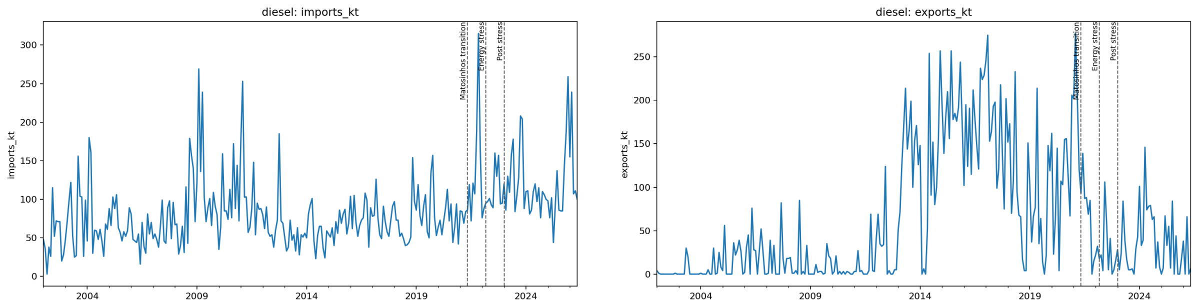

Figure 6: Monthly diesel imports (left) and exports (right).

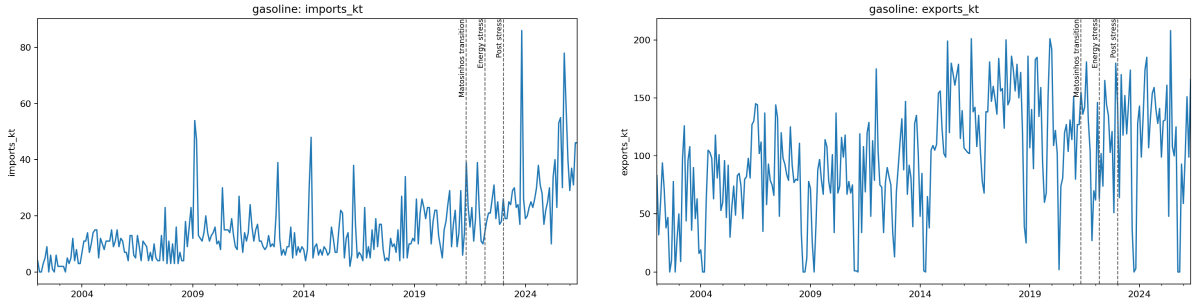

Figure 7: Monthly gasoline imports (left) and exports (right).

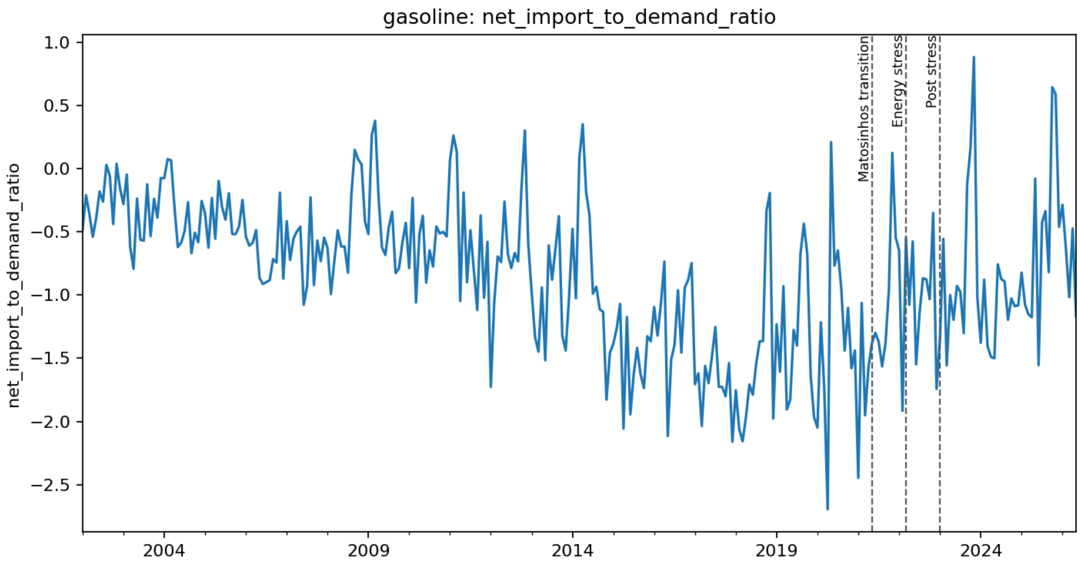

Figure 8: Monthly gasoline net imports to demand. The series stays far below zero throughout, consistent with the annual picture of a persistent export surplus.

Gasoline exports in the same year are unremarkable, at $z = -0.46$ and the 29th percentile, while gasoline imports are elevated at $z = 2.06$. Diesel demand in 2022 is close to baseline ($z = 0.41$), so the import surge is not demand-driven.

The stress is therefore diesel-specific on the supply side. That is consistent with the documented suspension of Russian oil-product and vacuum gas oil purchases feeding diesel manufacturing at Sines, and inconsistent with a general refining shutdown, which would have moved gasoline equally. The metrics are reported across three baseline windows and the ranking is unchanged in each.

8 Prices

8.1 Diagnostics and model choice

Weekly pre-tax prices are the primary measure. Augmented Dickey–Fuller tests on the levels fail to reject a unit root for all four series: Portuguese diesel $p = 0.124$, Spanish diesel $p = 0.242$, Portuguese gasoline $p = 0.119$, Spanish gasoline $p = 0.113$. A regression in levels is therefore at risk of being spurious for both products, and the model family is selected accordingly.

For diesel the pair is *not* cointegrated (Engle–Granger $p = 0.240$), so the short-run log-difference model carries the inference. For gasoline the pair *is* cointegrated ($p = 0.000026$), so an error-correction model is indicated and estimated.

8.2 The levels model, and why it is not used

The levels regression was estimated and is retained in the results archive, flagged as not valid for inference for both products. It is reported here because discarding it silently would hide how much specification choice matters, not because it supports any claim.

On that specification the diesel interaction term ES $\times$ post is -0.102 ($p < 0.001$): Portuguese prices appear to become *less* responsive to Spanish prices after the transition. The licensed short-run specification gives $+0.272$ ($p < 0.001$), the opposite sign. The levels model also reports a deterministic trend of -0.015 per week ($p = 0.005$). Both features are what a regression between two integrated series without a cointegrating relationship is expected to produce. No conclusion in this paper rests on it.

8.3 Short-run pass-through

Table 9: Short-run pass-through, $\Delta \log$ pre-tax price, HAC(8), $n = 1{,}077$ weeks.

Product	Term	Estimate	Std. error	p
diesel	$\Delta \log$ ES	0.7348	0.0341	< 0.001
	$\Delta \log$ ES $\times$ post	0.2719	0.0541	< 0.001
	post	-0.0004	0.0009	0.647
gasoline	$\Delta \log$ ES	0.7997	0.0593	< 0.001
	$\Delta \log$ ES $\times$ post	0.0352	0.1375	0.798
	post	-0.0003	0.0011	0.804

Weekly pass-through from Spanish to Portuguese pre-tax diesel prices rises from 0.73 before the May 2021 transition to approximately 1.01 after it, and the increase is precisely estimated. The gasoline interaction is small and statistically indistinguishable from no change.

8.4 Gasoline error correction

For gasoline the short-run coefficient is not the whole result. Table 10 reports the two-step Engle–Granger model the cointegration diagnostic indicates. The long-run relation is close to unit elastic, $\log \mathrm{PT} = 0.147 + 0.974 \log \mathrm{ES}$, and the speed at which Portuguese gasoline prices close a gap against that relation roughly triples after the transition, from -0.091 to -0.299 per week. The implied half-life of a deviation falls from about eight weeks to a little over two.

Table 10: Gasoline error-correction model, two-step Engle–Granger, HAC(8), $n = 1{,}077$ weeks.

Term	Estimate	Std. error	p
Disequilibrium (lagged)	-0.0907	0.0354	0.010
$\times$ post	-0.2080	0.0720	0.004
$\Delta \log$ ES	0.7889	0.0720	< 0.001
$\times$ post	0.1970	0.1648	0.232
post	-0.0019	0.0014	0.180

Gasoline linkage to Spain therefore did tighten. It tightened in the long-run adjustment channel rather than in contemporaneous pass-through, which a difference-only model cannot observe by construction. Reporting only the short-run coefficient would have supported the conclusion that gasoline was unaffected, and that conclusion would have been an artefact of the specification.

The disequilibrium term is a generated regressor, so the second-stage standard errors are conditional on the first-stage cointegrating estimate. HAC covariance mitigates but does not remove this.

8.5 Price spreads

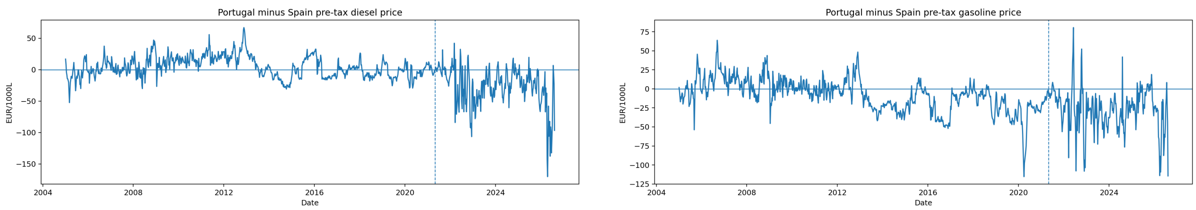

Figure 9: Portugal and Spain weekly pre-tax prices and their spread: diesel (left), gasoline (right).

The mean Portugal-minus-Spain pre-tax diesel spread moves from $+4.12$ to -26.66 EUR per 1000 litres across the transition, a change of -30.78; the gasoline spread moves by -18.80. These are descriptive.

Neither spread is stationary: diesel ADF $p = 0.605$, gasoline $p = 0.066$, both on 1,078 observations. Neither can therefore be treated as mean-reverting around a stable equilibrium, and spread levels must not be read as a disequilibrium measure.

9 Portugal–Spain Comparison

9.1 The physical divergence is Portugal-specific

Spain refines the same products, buys the same crude grades and was exposed to the same European energy shock in 2022. Its diesel balance therefore bounds how much of Portugal's change a common shock can explain. Table 11 shows that it explains very little.

Table 11: Diesel balance ratios, Portugal against Spain.

	Refinery output / demand			Net imports / demand		
Year	PT	ES	PT−ES	PT	ES	PT−ES
2018	1.049	0.899	+0.150	−0.056	−0.046	−0.010
2019	0.905	0.902	+0.003	+0.070	−0.032	+0.102
2020	1.077	0.932	+0.145	−0.084	−0.051	−0.033
2021	0.912	0.847	+0.065	+0.026	+0.009	+0.016
2022	0.867	0.904	−0.037	+0.140	+0.001	+0.139
2023	0.678	0.923	−0.245	+0.262	−0.041	+0.303
2024	0.855	0.898	−0.043	+0.105	−0.040	+0.145

Spanish diesel output covers between 0.85 and 0.93 of Spanish demand in every year shown, with no trend: Spain absorbed 2022 without a change in its physical balance. Portugal covered more of its own diesel demand than Spain did in 2018, 2020 and 2021, and less from 2022, falling to 0.678 in 2023 while Spain sat at 0.923. Spanish net imports stay within ±0.05 of balance throughout; Portugal reaches 0.262 in 2023.

This does not identify a causal effect, and Spain is not used as a control. What it does is constrain one competing explanation: a shock common to both countries cannot account for a divergence that appears in only one of them. Demand recovery, Sines maintenance and the Russian VGO disruption remain live alternatives, and the VGO disruption in particular was not common to Spain in the same degree.

9.2 Balance residuals

The Eurostat panel permits an internal consistency check absent from JODI. Defining the residual as $Q^{\text{refinery}} + M - X - D$, Spanish gasoline residuals run at 6–15 per cent of demand, reflecting stock changes, refinery fuel and statistical differences that the four reported terms do not close. Balance residuals are therefore reported but not interpreted as flows, and no claim in this paper is derived from a residual.

10 Robustness

10.1 Event windows

The pre/post comparisons use five pre-event and three post-event years. Repeating them over three window pairs leaves every conclusion unchanged.

Table 12: Pre/post difference around 2021 by window, 2021 excluded.

Product	Outcome	3 pre / 2 post	5 pre / 3 post	8 pre / 3 post
diesel	net imports / demand	+0.229	+0.275	+0.324
diesel	refinery output / demand	−0.242	−0.294	−0.348
gasoline	net imports / demand	+0.380	+0.433	+0.336
gasoline	refinery output / demand	−0.370	−0.420	−0.314

Every estimate keeps its sign across all three windows. The diesel ratios grow monotonically with a longer pre-window, which is expected: a longer baseline reaches further into the pre-hydrocracker regime and widens the measured gap. The gasoline estimates are flat to slightly smaller. No conclusion here depends on the window choice.

On the headline five-and-three window, diesel gross import dependence rises from 0.190 to 0.274, a change of 8.4 percentage points; the diesel net-import-to-demand ratio rises from −0.119 to 0.183, a change of 30.2 percentage points; and diesel refinery output coverage falls from 1.088 to 0.767. These are descriptive pre/post differences, not effect estimates.

10.2 Multiplicity

Sixteen Chow tests are reported and Benjamini–Hochberg adjustment is applied across all of them, not to a selected subset. Three survive at conventional levels. The monthly models are not included in that adjustment and their p-values should be read as unadjusted; the phase terms that carry the conclusions have $p < 0.005$, comfortably inside any reasonable correction for the twelve terms reported.

11 Limitations

1. **2022 confounds the post-closure window.** The Matosinhos transition and the European energy shock fall fourteen months apart. The monthly model separates them statistically, but COVID-recovery demand, Sines maintenance schedules and the Russian VGO disruption are competing explanations that are not separately identified.
2. **No causal design.** No control group, instrument or counterfactual is used. Every post-2021 result is an association measured around a documented event. The Spain comparison constrains the common-shock explanation but does not license difference-in-differences: no pre-trend or common-shock diagnostics have been estimated.
3. **DGEG trade covers only 2019–2024.** The 2013 break is corroborated against Eurostat, which is compiled from the DGEG submission rather than independent of it.
4. **Low annual power after 2021.** Three post-event annual observations cannot support a Chow test. The interrupted-trend models carry the annual evidence for 2022, and they impose a common trend that the Chow design does not.
5. **Generated regressor in the gasoline ECM.** The disequilibrium term is estimated in a first stage, so the reported second-stage standard errors are conditional on it. A single-step or maximum-likelihood estimator would price that uncertainty properly.
6. **JODI assessment codes unverified.** All Portuguese observations carry assessment code 1, whose exact semantics have not been confirmed against JODI documentation.
7. **Retail, not wholesale, prices.** The Weekly Oil Bulletin reports consumer prices. Pass-through results describe retail co-movement, not refinery-gate or international wholesale transmission.
8. **Balance residuals are not closed.** The four Eurostat terms do not sum to zero and stock changes are not modelled.

12 Conclusions

The 2013 Sines hydrocracker produced an unambiguous structural change. Diesel refinery output and exports broke upward at 2013 and Portugal moved from net diesel importer to net diesel exporter, a result that survives multiplicity adjustment across sixteen tests and is corroborated by interrupted-trend models with level shifts of +1,900 kt in exports and −0.52 in the net-import ratio.

After May 2021 the direction reverses. By 2023 domestic diesel manufacturing covered two thirds of demand, the lowest coverage in twenty years, and imports reached a series high. Two annual designs and one monthly design agree on the direction; they disagree only on whether three annual post-event observations can support a Chow test, which they cannot. The monthly design separates the closure from the 2022 energy shock and finds both independently associated with lower refinery output and higher net import dependence, the 2022 effect being the larger.

Spain, refining the same products under the same European shock, shows no comparable movement. That constrains but does not eliminate the common-shock explanation.

On prices, short-run pass-through from Spanish to Portuguese pre-tax diesel prices rises from 0.73 to approximately 1.01. Gasoline contemporaneous pass-through is unchanged, but its error-correction speed roughly triples, so both fuels became more tightly linked to Spanish prices through different channels. That contrast is only visible because the model family was selected from diagnostics rather than assumed.

This is an association, not a demonstrated causal effect. The closure and the largest European energy shock in decades fall within fourteen months of one another, and this design cannot separate their contributions to the post-2021 level. What the evidence supports is that Portugal's diesel system moved from surplus to deficit over the study window, that the movement is concentrated around two documented events, that it is not shared by Spain, and that Portuguese fuel prices now track Spanish prices considerably more closely than before.

Claim–Evidence Matrix

Claim	Where shown	Level	Remaining caveat
Sines hydrocracker entered commercial production in 2013	Section 1	documented event	Corporate disclosure, not outcome data
Matosinhos refining ceased May 2021	Section 1	documented event	Announcement date differs from closure date
2022 Russian product and VGO disruption	Section 1	documented event	Establishes timing, not magnitude
Diesel demand averaged 5,064 kt, output 4,745 kt	Section 4.1	descriptive	Window means, not a model
Portugal was a net diesel exporter 2013–2020	Tables 2, 3	descriptive	Ratio is not self-sufficiency
Gasoline is a net export product in every year	Tables 2, 3	descriptive	Same caveat
Balance at each event year	Table 4	descriptive	2021 marked transition
Refining regime by year	Table 4	documented	Regime label, not capacity
Diesel net-import ratio breaks at 2013	Table 5	structural break	Diagnostic, not a causal estimator
No Chow break detected at 2022	Table 5	structural break	Three post-event years; low power
Interrupted trends do detect 2022	Table 6	association	Imposes a common trend
Monthly phase means shift after May 2021	Table 7	descriptive	No seasonality control
Refinery output falls 105 kt/month at the closure boundary	Table 8	association	Quote with its phase-trend term
2022 stress adds a further −152 kt/month	Table 8	association	Confounded with demand recovery

Claim	Where shown	Level	Remaining caveat
Monthly panel spans 293 modelled months	Section 2.2	descriptive	Panel extends beyond the window
2022 diesel exports at the 0th baseline percentile	Section 7	descriptive	Baseline windows chosen ex ante
All four price levels are non-stationary	Section 8.1	descriptive	ADF only; no break-robust unit-root test
Diesel not cointegrated, gasoline cointegrated	Section 8.1	descriptive	Engle–Granger, single specification
Levels model gives the opposite sign and is not used	Section 8.2	—	Retained as a specification warning only
Diesel pass-through rises 0.73 to 1.01	Table 9	association	Retail, not wholesale, prices
Gasoline short-run pass-through unchanged	Table 9	association	Not the whole picture; see the ECM
Gasoline adjustment speed triples	Table 10	association	Generated-regressor standard errors
PT–ES diesel spread moves -30.8 EUR/1000L	Section 8.5	descriptive	Spread is non-stationary
Neither spread is mean-reverting	Section 8.5	descriptive	Gasoline marginal at $p = 0.066$
Spanish diesel balance is flat while Portugal's falls	Table 11	descriptive	Constrains, does not exclude, common shocks
Eurostat balance residuals are not closed	Section 9.2	descriptive	Stock changes not modelled
Pre/post differences hold across three windows	Table 12	descriptive	Magnitudes vary with baseline length
Headline pre/post dependence change	Section 10.1	descriptive	Not an effect estimate
JODI coverage is complete for every year	Table 1	descriptive	Assessment-code semantics unverified
Sources agree on 89% of full-window trade cells	Section 2.5	descriptive	Eurostat is DGEG-derived
Sources and retrieval points are frozen	Section 2.1	documented	Snapshot of one vintage
Hypotheses were pre-specified	Section 1	documented	Recorded before estimation
No causal effect of the closure is claimed	—	—	2022 is not separable by this design

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Appendix A. Data and Code Availability

All data underlying this paper are public and are listed in the References. The analysis code, the derived tables behind every figure and table above, and a document recording the provenance of each one are available in the project repository, reference [9].

Derived results are archived with a checksum manifest that accounts for every file in the archive, so the tables and figures reproduced here can be verified against the exact inputs used to produce them. Eight automated readiness checks are applied before any figure is reported: they verify archive completeness, annual coverage of the monthly source, validity of the monthly panel and its models, availability of the cross-country balance, completeness of the price diagnostics including the recorded model-family verdict, and source reconciliation. The reconciliation check additionally requires every out-of-tolerance cell to be named with a reason, so a later data vintage cannot widen the gap between sources silently. All eight pass for this paper.